
Task 5: Exploratory Data Analysis (EDA)

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Tools used:

- Visual Studio code (vs code)
- Python
- Pandas
- Matplotlib
- Seaborn

Objective

The objective of this task is to perform Exploratory Data Analysis (EDA) on the Superstore Sales dataset to identify patterns, trends, relationships, and anomalies using statistical methods and visualizations.

Dataset: Sample Superstore Dataset

Data Exploration

Dataset Information

- Loaded the dataset successfully.
- Used `df.info()` to view the dataset structure.
- Used `df.describe()` to generate summary statistics.
- Used `df.isnull().sum()` to check for missing values.
- Used `value_counts()` to count records in each category.

Observation

- The dataset contains sales records of different products.
 - No significant missing values were found.
 - Categories include Furniture, Office Supplies, and Technology.
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Visualizations and Observations

1. Histogram (Sales Distribution)

Observation:

The histogram shows that most sales are of smaller amounts, while only a few transactions have very high sales. The distribution is right-skewed.

2. Box Plot (Profit)

Observation:

The box plot reveals several outliers in the Profit column. Some transactions have unusually high profits, while a few have significant losses.

3. Scatter Plot (Sales vs Profit)

Observation:

Higher sales do not always result in higher profits. Some high-sales transactions generated low or negative profit.

4. Correlation Heatmap

Observation:

The heatmap shows the relationship between numerical variables. Sales and Profit have a moderate positive correlation, while Discount has a negative correlation with Profit.

5. Pair Plot

Observation:

The pair plot displays pairwise relationships among Sales, Profit, Quantity, and Discount. It helps identify trends, distributions, and outliers.

Summary of Findings

- The dataset contains valuable sales information across multiple product categories.
- Most sales are concentrated at lower values.
- Profit contains several outliers.
- Sales and Profit have a moderate positive relationship.
- Discount negatively impacts Profit.
- Exploratory Data Analysis helped identify patterns, trends, and anomalies in the dataset.

Conclusion

Exploratory Data Analysis was successfully performed using Python libraries such as Pandas, Matplotlib, and Seaborn. The analysis provided useful insights into sales performance, profit trends, and relationships between variables, making the dataset ready for further analysis and machine learning.
